EXCLUSIVE

# IAS PRELIMS 2026

# CURRENT AFFAIRS

GS-1 (DYNAMIC)

# REVISION

MODULE - 17

## SCIENCE & TECHNOLOGY-4

# What are the Space Sector Reforms in recent times?

*The government has also approved setting up of Rs. 1,000 crore venture capital fund dedicated to the space sector under IN-SPACe for the coming five years.*

- The Space Sector Reforms in 2020 allows participation of Indian private sector in space activities.
- As a part of these reforms, government has delineated the roles of various entities i.e., IN-SPACe, ISRO and NSIL.
- The government released Indian Space Policy 2023, that provides a level playing field to Non- Government Entities (NGEs) in the space sector by enabling their participation across the entire value chain of space activities.
- Amendments were made to the Foreign Direct Investment Policy for Space Sector.
  - ✓ Upto 74% under Automatic route: Satellites-Manufacturing & Operation, Satellite Data Products and Ground Segment & User Segment. Beyond 74% these activities are under government route.
  - ✓ Upto 49% under Automatic route: Launch Vehicles and associated systems or subsystems, Creation of Spaceports for launching and receiving Spacecraft. Beyond 49% these activities are under government route.
  - ✓ Upto 100% under Automatic route: Manufacturing of components and systems/ sub-systems for satellites, ground segment and user segment.

# What do you understand by Integrated Air Drop Test (IADT)-1?

- Recently, ISRO successfully carried out its first Integrated Air Drop Test (IADT)-1, a crucial milestone in the preparation for the country's made-in-human space flight mission, Gaganyaan.
- For IADT-1, an Indian Air Force Chinook helicopter lifted a 4.8-ton dummy crew module into the air. At the designated altitude, helicopter released the capsule. From then on, automated systems triggered the sequential deployment of parachutes that facilitated these parachutes to splash down safely.
- The IADT is designed to evaluate the parachute-based deceleration system that will bring the Gaganyaan crew module down safely after reentry.

# What is the significance of Group Captain Shubhanshu Shukla's space voyage?

- Group Captain Shubhanshu Shukla was back on Earth. The capsule Crew Dragon, "Grace" splashed down in the Pacific Ocean off the California coast.
- Shukla circled the world 288 times from about 400 km above, clocking over 12.2 million km and he became the most traveled Indian.
- The US space agency, NASA, without a fee, entrusted him the crucial task of piloting the capsule.
- Shukla also performed about 60 experiments, under conditions of microgravity, under seven India specific heads of research, such as checking muscular dystrophy with electrical pulses on the body of an Indian, seed germination of Indian strains of crops like millets and mustard and the survival of Indian strains of tardigrades and cyanobacteria.

*India spent about $70 million to train Shukla and send him to the ISS.*

# What do you know about the Third Launch pad at the Satish Dhawan Space Centre (SDSC) in Sriharikota?

- The Union Cabinet approved the construction of a Third launchpad at the Satish Dhawan Space Centre (SDSC) in Sriharikota. The launchpad will have an outlay of Rs. 3,984.86 crore and is targeted to be completed by early 2029.
- The launchpad comes as the Indian Space Research Organization (ISRO) seeks to launch its Next Generation Launch Vehicles (NGLVs) starting in 2031.
- It will be configured as universal and adaptable as possible that can support not only NGLV but also the Launch Vehicle Mark III vehicles with semi cryogenic range as well as a scaled-up configurations of NGLV.

*The previous launchpad has been operational for almost two decades and this new pad is expected to boost ISRO's launch capabilities and capacities in Sriharikota.*

# What do you know about the Germline Modifications and Somatic Cell Editing?

- Germline modifications are the genetic changes that are introduced in the germ cells (sperm and eggs). These changes can be passed on to future generations.
- A person's genome can also be edited before birth at the early embryonic stage. Genome editing technologies like CRISPR can be used to edit the genome of the embryos. In 2018, CRISPR was used to edit embryos in China to make them resistant to HIV.
- Somatic cells editing involves editing the cells of organs like brain, kidneys, heart, etc. These are not passed on to the subsequent generations.

# What do you understand by Mitochondrial Diseases? What are the technological advances to prevent Mitochondrial Diseases?


- Mitochondrial diseases are a group of disorders caused by dysfunctional mitochondria, the energy producing structures within cells. These diseases are maternally inherited because mitochondria in the offspring come exclusively from the Mother's egg.
- There are Mitochondrial Replacement Therapies (MRTs). MRTS are designed to prevent the transference of mitochondrial diseases from mother to child. Some examples are maternal spindle transfer, pronuclear transfer.

# What do you know about Blockchain Technology?


- Blockchain technology was initially developed to support Bitcoin, and now its applications have expanded far beyond that.
- It is being used for various purposes such as supply chain management, voting systems, identity verification, smart contracts, etc.
- This is a decentralized ledger that records all transactions across a network.
- It is a public ledger that everyone can inspect, but which no single user controls.
- This high level of transparency allows for trust and accountability in transactions.
- Its core features of decentralization, transparency, and immutability make it suitable for a wide range of applications.

# What do you know about Anusandhan National Research Foundation (ANRF)?

- ANRF (Anusandhan National Research Foundation) has been established by ANRF Act, 2023.
- The provisions of the ANRF Act, 2023 came into force on 5th February, 2024.
- ANRF main goal is to unleash Indian research and innovation talent to achieve global scientific and technological excellence.
- The first meeting of the Executive Council of ANRF was held under the Chairmanship of Prof. Ajay K. Sood, Principal Scientific Adviser to the Government of India on August 22, 2024.
- Afterwards, Prime Minister Shri Narendra Modi chaired the first meeting of the Governing Board of Anusandhan National Research Foundation on 10th September, 2024.

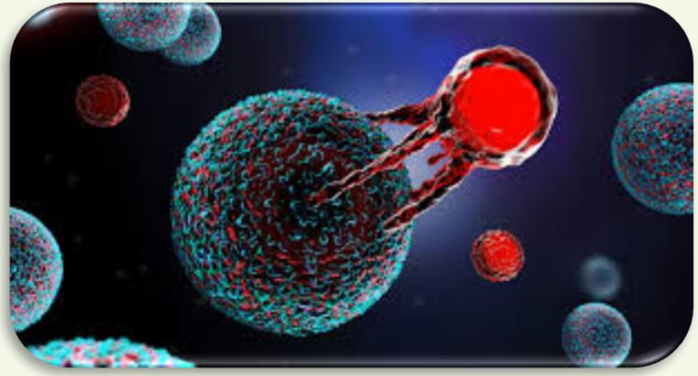
# What do you know about the National Mission on Interdisciplinary Cyber Physical System?


- The National Mission on Interdisciplinary Cyber Physical System (NM-ICPS), aims at development of technology platforms to carry out R&D, translational research, product development, incubating and supporting startups as well as commercialization.
- Total 25 Technology Innovation Hubs (TIHs) have been established in the areas of advanced technologies which includes:
  - ✓ Artificial Intelligence and Machine Learning (AI/ML).
  - ✓ Robotics.
  - ✓ Cyber Security.
  - ✓ Data Analytics & Predictive Technologies.
  - ✓ Technologies for Agriculture & Water.
  - ✓ Technologies for Mining.
  - ✓ Advanced Communication Systems.
  - ✓ Quantum Technologies.
- Each TIH is created as a Section-8 Company, an independent entity within the Host Institute with the involvement of industry as potential members for co-development, partnerships and for commercialization.

# What do you know about the Software Technology Parks and Global Capability Centers?

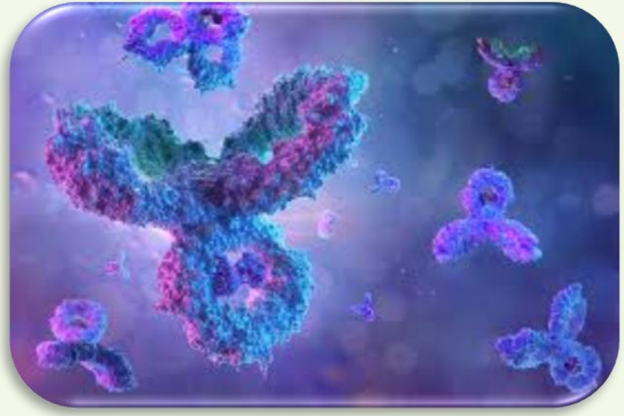

- Under the Software Technology Park (STP) Scheme, STPs have been established in 65 cities across India, with 57 centres in Tier-2 and Tier-3 cities. They provide incubator facility which help entrepreneurs to convert their innovative ideas into startups.
- The incubator facility provides opportunity of meeting with investors like Venture Capitalists, mentors from IITs, NITs, etc.
- Through Global Capability Centers, products like vehicle engines, semiconductors chips, big machineries are getting designed in India. At present, India hosts around 1,800 GCCs employing 19 lakh persons.

# What is the process involved in CAR T-cell Therapy?



- Many harmful cells have protein markers called antigens on their surface.
- T-cells in a patient's immune system are extracted, genetically tweaked to possess Chimeric Antigen Receptors (the CAR in CAR-T) that can spot these antigens and then return to the patient's body.
- Several CAR T-cell therapies are licensed to treat Leukemia and Lymphoma. Here, the mechanism is that T-cells engineered with CARs can recognize a protein called CD19, which is found on the outside of those cancer cells.

# What are Monoclonal Antibodies (mAbs)?



- Monoclonal Antibodies (mAbs) are proteins made in laboratories that act like proteins called antibodies in our bodies.
- The word "monoclonal" refers to the fact that the antibodies created in the laboratory are clones. They are exact copies of one antibody.
- The generic names of the products often include the letters "mab" at the end of the name.
- These are used for diagnosis, disease treatment and research:
  - ✓ As probes to identify materials in laboratories or for use in home testing kits like those for pregnancy or ovulation.
  - ✓ For diagnosis.
  - ✓ For disease treatment.

***In USA, approvals of monoclonal antibody therapies has been raising since the first monoclonal antibody drug for humans was approved in 1986.***

# WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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